2023-01-0858 Published 11 Apr 2023

Comparison of Infrastructure- and Onboard Vehicle-Based Sensor Systems in Measuring Operational Safety Assessment (OSA) Metrics

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Citation: Das, S., Rath, P., Lu, D., Smith, T. et al., "Comparison of Infrastructure- and Onboard Vehicle-Based Sensor Systems in Measuring Operational Safety Assessment (OSA) Metrics," SAE Technical Paper 2023-01-0858, 2023, doi:10.4271/2023-01-0858.

Received: 08 Nov 2022 Revised: 11 Jan 2023 Accepted: 06 Feb 2023

Abstract

The operational safety of Automated Driving System (ADS)-Operated Vehicles (AVs) are a rising concern with the deployment of AVs as prototypes being tested and also in commercial deployment. The robustness of safety evaluation systems is essential in determining the operational safety of AVs as they interact with human-driven vehicles. Extending upon earlier works of the Institute of Automated Mobility (IAM) that have explored the Operational Safety Assessment (OSA) metrics and infrastructure-based safety monitoring systems, in this work, we compare the performance of an infrastructure-based Light Detection And Ranging (LIDAR) system to an onboard vehicle-based LIDAR system in testing at the Maricopa County Department of Transportation SMARTDrive testbed in Anthem, Arizona. The sensor modalities are located in infrastructure and onboard the test vehicles, including LIDAR, cameras, a real-time differential GPS, and a drone with a camera. Bespoke localization and tracking algorithms are created for the LIDAR and cameras. In total, there are 26 different scenarios of the test vehicles navigating the testbed intersection; for this work, we are only considering car following scenarios. The LIDAR data collected from the infrastructure-based and onboard vehicle-based sensors system are used to perform object detection and multi-target tracking to estimate the velocity and position information of the test vehicles and use these values to compute OSA metrics. The comparison of the performance of the two systems involves the localization and tracking errors in calculating the position and the velocity of the subject vehicle, with the real-time differential GPS data serving as ground truth for velocity comparison and tracking results from the drone for OSA metrics comparison.

Introduction

It is anticipated that the future of mobility could revolve around a smart infrastructure capable of real-time monitoring of traffic and road conditions, two-way communication with vehicles and vulnerable road users (VRUs), and promoting safer and more efficient transportation. Smart infrastructure is not in widespread use yet and companies currently working in the Automated Driving System (ADS)-Operated Vehicle (AV) space primarily rely on onboard sensors on the vehicle alone. However, infrastructure-based systems deployed for smart cities' safety and security purposes could also be used for monitoring of AVs by infrastructure owner-operators (IOOs). It is important to understand the measurement uncertainty of measurements taken by the infrastructure-based sensors, regardless of how the measurements are used. One definite advantage of infrastructure based LIDAR is the height of placement of LIDAR which gives a larger field of view compared to LIDAR mounted on vehicle top.

Previous work in the IAM team by Anshuman et al. [1] has shown how the infrastructure-based LIDAR can be used for monitoring and collecting measurements of vehicle position and velocity that can be used to calculate operational safety assessment (OSA) metrics introduced by Wishart et al. [2] that can be used in assessing the operational safety of AVs. In this work, we investigate the performance of infrastructure-based LIDAR system in comparison to onboard vehicle-based LIDAR monitoring system. This work is intended to answer the following questions: (1) How is infrastructure based monitoring system compared to an onboard vehicle-based system in terms of performance and accuracy? (2) Will an infrastructure-based system benefit automated vehicles in a connected space?

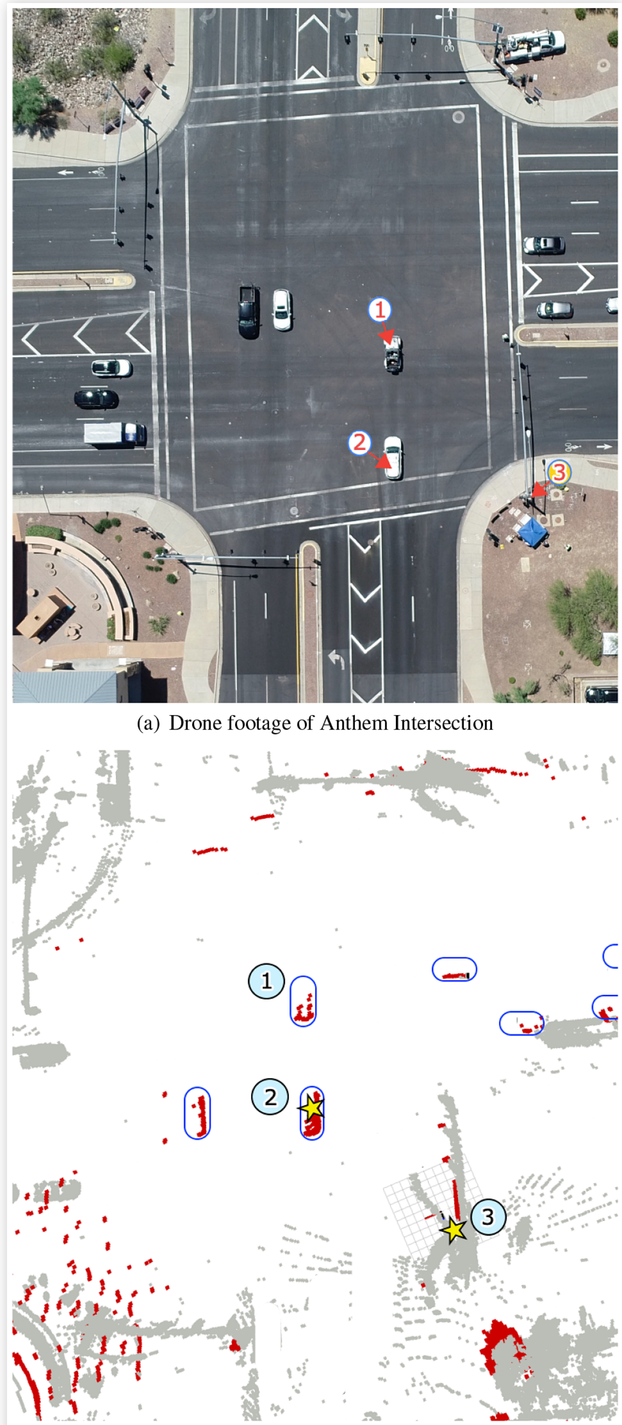
In this study, we investigate the localization and velocity tracking results from onboard vehicle LIDAR and infrastructure-based LIDAR and compare those with that of drone (camera) serving as ground truth for the study. Finally, we compute a safety metric, namely time-to-collision (TTC) from all the sensor modalities mentioned and compare those to understand the performance and reliability of those two methods.

Related Work

In recent years numerous efforts have gone into object detection and tracking of LIDAR-based point cloud data. There has been a shift from traditional clustering/segmentation-based methods to machine learning and deep learning-based methods for detecting objects in point cloud data. The ability of convolutional neural networks (CNN) to take advantage of spatial and local correlations in grid-like data has led to their widespread use in computer vision, especially in image processing. Point clouds, on the other hand, have a highly sparse distribution. Pattern recognition is difficult owing to the erratic structure and geometry of point cloud data. As a result, convolving directly on point clouds result in extreme representational distortion [3]. VoxelNet [4] converts point clouds to volumetric grids and utilizes a 3D CNN for each spatial dimension. A similar approach is used by SECOND [5]. However, such approaches have memory growth in the order of $O(N^3)$ and hence lead to huge compute overheads with humongous memory footprints.

A common strategy/preprocessing step for detection of objects in point clouds is to convert the 3D data into 2D Bird Eye View (BEV) data by compacting a dimension (usually along the z-axis). Literature has several papers utilizing this strategy for 3D object detection. PointPillars [6] is one of such approaches that projects voxels along pillar like array along the BEV axis. This however, leads to resolution loss along the projection axis which becomes a bottleneck for representational learning. Falling back on voxel based approaches, Voxel-RCNN [7] accumulates features from voxels using 3D convolutions for proposal improvement. SE-SSD [8] utilizes a teacher network to supervise a student network using distilled knowledge transfer between the networks. In our work we use Complex-YOLO by M. Simon et al. in Complex YOLO [9]. By incorporating an imaginary and a real fraction into the regression network, Complex-YOLO uses an Euler-Region-Proposal Network (E-RPN) to predict the heading of the objects.

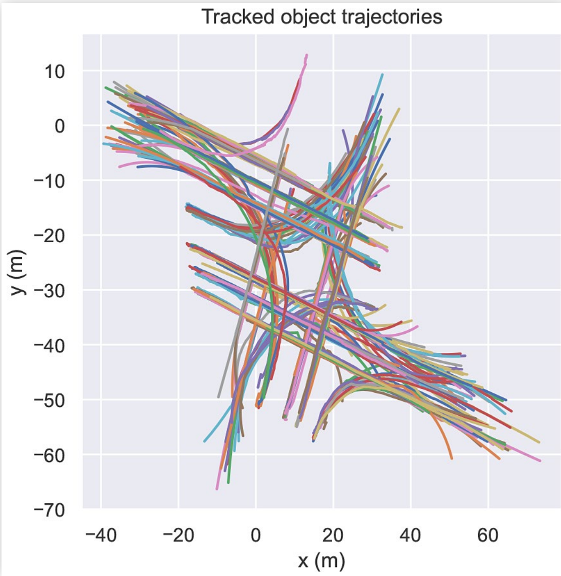
FIGURE 1 In the above two figures marker 1 indicates the lead vehicle, marker 2 indicates the following vehicle and marker 3 indicates the position of infrastructure-based LIDAR



(a) Drone footage of Anthem Intersection

(b) Point cloud map of anthem intersection withinfrastructure LIDAR point cloud superimposed

FIGURE 2 Trajectory of tracked objects



Tracking of vehicles is essential for getting the velocities after detection phase. Reid et. al published the Multiple Hypothesis Tracking (MHT) [10] algorithm which is considered as the earliest successful implementation of multi-object target tracking. Target states for each object is predicted using Kalman Filters and data association is formulated as joint hypothesis. New measurements are used to propagate the filter, and hypothesis with lower probabilities are terminated to reduce the number of possible associations. For track termination, MHT individually prunes out hypothesis with low probability. Following the ideas from MHT, a relatively recent work by Chanho et al. [11] uses a least square based online appearance modelling approach for hypothesis tracking. The algorithm utilizes a weighted combination of conventional hypothesis score along with a appearance score for each track. The approach has proven to have generated impressive results on benchmark datasets.

Another very popular approach to multi-object tracking is Joint Probabilistic Data Association (JPDA) [12]. A marginal probability on the joint data association space is calculated by the algorithm. These are made up of all conceivable pairings of measurements to targets where each measurement (apart from the dummy measurement associated with incorrect or missing measurements) is allocated to only one target, and each target is assigned to a single measurement exclusively. JDPA can be enhanced by taking multiple frames to consideration. This, however, makes the combinatorial explosion worse even with the use of gating strategies, hence it is hardly ever utilized in real-time applications. Inspired with the simplistic formulation of JPDA, Seyed et. al proposed m-JPDA [13] and formulated the data association problem as a linear program that could be solved using LP-relaxation [14]. With reduced association complexity, their 3 frame JPDA with 100 highest probability hypothesis has shown performance comparable to state of the art tracking algorithms.

Other recent approaches in the literature include cost-flow based [15], markovian monte carlo [16], heuristic [17] algorithms. Many of such approaches trade off accuracy with complexity and hence it is impractical to use them for real-time tracking purposes. Advancements in machine learning and deep learning has aided real-time multi-object tracking. Alex et. al have proposed Simple online and real-time tracking (SORT) [18] that utilizes a CNN for object re-identification. The appearance features from the CNN have proven to be useful for detection after long term occlusion. SORT also utilizes a simple Kalman filter with constant velocity model which has a weighted contribution to the association problem along with the appearance features. The authors published another follow-up paper that uses a deeper network architecture to reduce identity switches. Another recent confidence tracking algorithm [19] that utilizes recursive cluster associations based on global pose transformations and point cloud change detection has shown stable tracking performance in real time scenarios.

Methodology

Our method includes three stages, those are preprocessing, detection, tracking, and lastly the safety metrics calculation. This section elaborates on each of these in detail.

Preprocessing

The YOLO [20] network is known for real-time object detection from 2D image is extended from 2D pictures to 3D point clouds by M. Simon et al. in Complex YOLO [9]. The main components of Complex YOLO are: (1) the transformation of a 3D point cloud to bird's eye view, (2) Complex YOLO on BEV map, and (3) 3D bounding box construction. In this section we are going to discuss creation of BEV map and the other two components are discussed in the next sub section (Detection).

The 3D point cloud is first turned into BEV by compacting the points along the sky-facing axis(Z-axis in our case). Since BEV image height and width is limited, we can consider point clouds only within certain limits, which is accomplished by filtering, and BEV is divided into a grid of cells, each cell representing a pixel that covers a region on the road surface. A color image in RGB format has 3 channels, i.e., R, G, B channels. In Complex YOLO [9], these three channels represent the height, density, and intensity of points in the point cloud.

The actual region considered for the method is in 50 m range, both along the longitudinal and the lateral directions. 50 m range is based on the original research, and design convention in existing Complex YOLO [9] implementations. The area is then divided into grid cells by defining the resolution of BEV image, and we adjusted the BEV image size to 608 x 608 pixels, resulting in a spatial resolution of approximately 8cm.

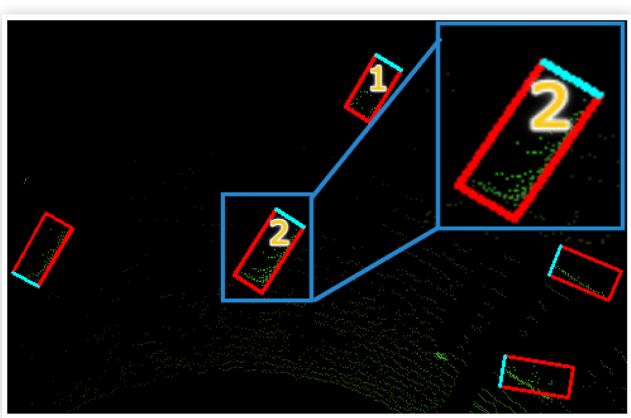
After the grid is created for the detection area, for each cell, we need to find the height, intensity, and density value. There could be multiple points in a single cell. In such cases the point at the topmost position in the z-axis(upward direction) is selected, removing other points in the cells for the height channel and the point with maximum intensity is chosen for the intensity channel. The density channel is created by normalizing the density of all points mapped into a cell. This gives us the BEV image that is consumed by the Complex YOLO network, which is discussed in the next section.

In addition to all the steps mentioned, the infrastructure-based LIDAR point cloud has a prior step before creating the BEV image. That is, the coordinate frame for the infrastructure-based LIDAR is first rotated so that X-Y plane is approximately parallel to the road surface and z-axis is normal to the road surface, and then origin of reference frame is translated to a landmark on ground. This is done for ease of understanding as well as filtering, as the default origin is at the position on the pole where the LIDAR was mounted and z-axis(head of LIDAR sensor) was tilted towards the ground.

Object Detection

After the BEV image is created, it is fed to the Complex YOLO for detection of vehicles. Although a reduced YOLOv2 [21] Convolution Neural Network(CNN) architecture was employed in the original Complex YOLO; in our implementation, we have used the YOLOv4 CNN architecture. In addition to the basic YOLO network used for object detection in 2D images, the Complex YOLO has a complex angle regression and Euler-Region Proposal Network (E-RPN) that serve to obtain the rotation of the bounding box to find the identified object better. The YOLO network is set up to anticipate 75 features by dividing the image into 16 x 32 grids. The YOLO model consists of 18 convolutional layers and five pooling layers. There are also three intermediary layers for feature restructuring. Every detection box has five parameters two-dimensional position in the BEV map, width and length, and two components for the orientation angle. There is another parameter that indicates whether the bounding box encompasses an actual object and is correctly placed. Finally, the Region Proposal Network employs five more parameters to predict precise object orientations and borders.

FIGURE 3 Detection result from infrastructure based LIDAR.



One distinguishing feature that makes Complex YOLO from other approaches is the addition of an orientation angle to the classical Grid RPN approach, which estimates only bounding box location and shape, which it encodes as a complex angle in Euler notation (hence the name "E-RPN") so that the orientation can be reconstructed. The regression values are then passed directly to the computation of a loss function, which is based on the YOLO concept (i.e., the sum of squared errors) but extends it by an Euler regression part, which is obtained by calculating the difference between the ground truth and predicted angle, which is always assumed to be inside the circle shown above.

Tracking and Data Association

For tracking vehicles across consecutive frames, the Extended Kalman Filter (EKF), which is the non-linear version of the well-known Kalman Filter, is employed in this work. Tracks are initialized with detection position, and a constant velocity motion model is assumed. For data association, Simple Nearest Neighbor (SNN) is used. SNN computes the Mahalanobis distances of all possible track and measurement pair combinations, then updates the closest association pair in an iterative way. The association complexity is decreased by the implementation of a gating method. Fig. 2 shows the tracked trajectories of all the vehicle that passed through the intersection.

Safety Metric

In this study, we are only considering the Time-to-Collision (TTC) safety metric. TTC is the time required for a crash to occur in a particular circumstance if both the lead and follower vehicles continue to travel at their current speeds. A metric violation occurs when the measured TTC drops below the predetermined threshold. TTC is calculated using the distance and the relative speed between the lead and follower vehicles, as provided in the equation below. Where X_L and X_F are the positions of the lead and the follower vehicle, X_L and X_F are the speed of the lead and follower vehicle, respectively. A negative TTC value indicates a more complex situation, while a zero TTC value indicates a crash.

$$TTC = \frac{X_L - X_F}{V_F - V_L}$$

Experiment and Data Collection

The data used in this study was collected at the Maricopa County Department of Transportation SMARTDrive testbed in Anthem, Arizona. It was only focused on only one intersection. We had sensors mounted on poles at the intersection, as well as on two vehicles that were used to study different scenarios about the intersection. Except these a drone was also used to capture bird's eye view(BEV) of the whole intersection via a camera attached to the drone. This study only focuses on the LIDAR data and uses drone tracking data as a pseudo ground truth for the base of comparison.

Onboard Vehicle Sensors

There were two vehicles - (1) one jeep from Exponent equipped with real-time differential GPS vehicle and (2) a car (Trident) from ASU Luminosity Lab, equipped with a Ouster OS1 64 channel LIDAR positioned on roof top-center and camera. In the scenario considered for this study the Trident car follows the jeep as it can be seen in the Figure 1, which is why the jeep is referred as lead vehicle and the car is referred as follower all through this paper.

Infrastructure-Based Sensors

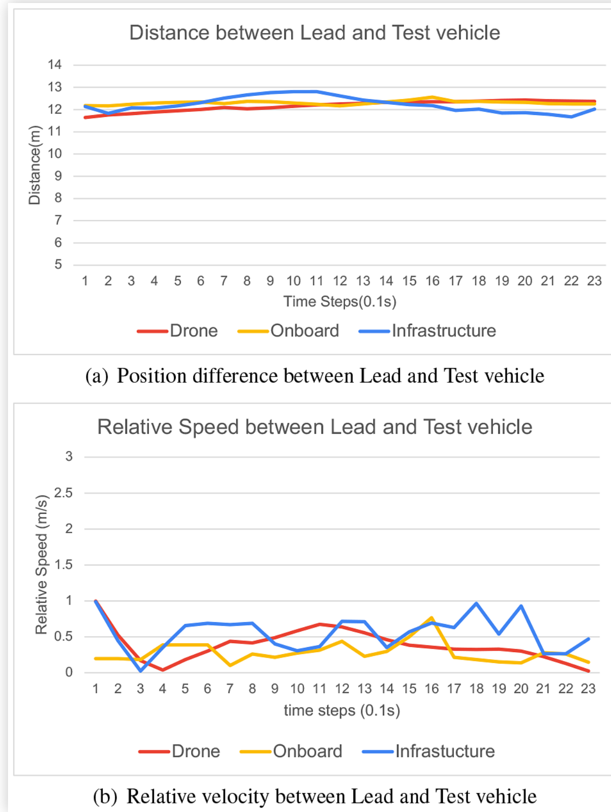
On the infrastructure side three Ouster OS1 LIDARs were mounted on three poles at the intersection in three different corners. Two LIDAR placed on opposite corners were set at an azimuth angle of 45° and one LIDAR was set with 90° azimuth angle. Cameras were also mounted to the poles in the data collection, but camera data is not used for this study.

Results

With the detection and tracking method as mentioned in 4, localization and speed of the lead and following vehicle were extracted from infrastructure-based LIDAR data and Onboard vehicle LIDAR data. The localization and speed measurement results were validated by considering the drone videos tracking results as the pseudo ground-truth or reference measurement. In the scenario under consideration, both the lead vehicle and follower vehicle could only be tracked for about 2.3s while in the intersection, as after 2.3s the point cloud of the lead vehicle almost disappears in the infrastructure based LIDAR, making it difficult to be detected by the algorithm. Frequency of LIDARs were 10Hz, therefore there are 23 frames captured in those 2.3s. The localization/position difference and the relative speed between the lead and the follower vehicle were computed for those 2.3s. However the frequency of drone tracking data was 30Hz, therefore we had to downsample the drone tracking data to 10Hz to ease the process of comparison. It should be noted that there was no time synchronization between all the three modalities under consideration here, thus there could be an error within 0.1 sec in time domain between one and another.

The overall result of localization difference from drone tracking, infrastructure based tracking and onboard vehicle based tracking are illustrated in the Figure 4(a) and the relative speed between the vehicles are illustrated in the Figure 4(b). From both of these figures, it can be observed that deviation in infrastructure based LIDAR results become prominent towards the last few frames, which is due to the scarcity of points as the vehicle move away from the LIDAR. This is more clearly observed in the Figure 5, in which the speed of the lead

FIGURE 4 Comparison of results from infrastructure based LIDAR and on-board vehicle LIDAR with drone as pseudo ground truth



(a) Position difference between Lead and Test vehicle

(b) Relative velocity between Lead and Test vehicle

FIGURE 5 Speed of Lead vehicle from LIDAR vs Drone vs GPS

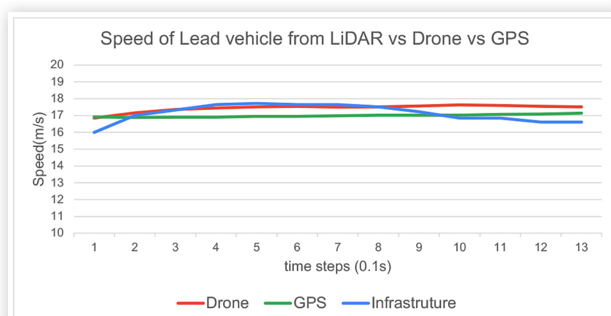
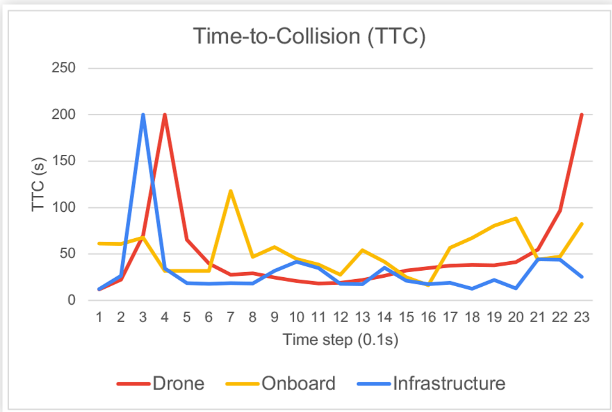


TABLE 1 Localization and speed measurement error using the Drone results as the reference.

Sensor Modality	Average Localization Error (m)	Maximum Localization Error (m)	Average Velocity Error (m/s)	Maximum Velocity Error (m/s)
Infrastructure LIDAR	0.3840	0.7052	0.2547	0.6437
On-board Vehicle LIDAR	0.1928	0.5493	0.2312	0.7963

FIGURE 6 Time-to-Collision computer from different sensor modalities.



vehicle measured from the infrastructure LIDAR is compared with drone and GPS measurements.

Average error and maximum error in localization and speed measurement is provided in the Table 1. Figure 6 illustrates the plot of time-to-collision(TTC) calculated from the onboard LIDAR tracking and infrastructure LIDAR tracking compared to that calculated from drone tracking results. The outliers were cut-off after values above two hundred for better illustration of the box plot. Interesting thing to note here is that the TTC curve for infrastructure based LIDAR is closer to the TTC curve for the drone, although average localization error and average velocity error of infrastructure based tracking is higher than the onboard vehicle based tracking with respect to drone tracking.

From these results it can be said that the infrastructure LIDAR based tracking is as reliable as an onboard LIDAR based tracking, but the preference of one over other depends on the use case and application.

Conclusion and Future Work

In this paper, the measurement errors of infrastructure-based LIDAR measurements are compared to those of onboard vehicle LIDAR measurements, and their effect on the accuracy of OSA metrics calculation was investigated. The results extracted from the infrastructure-based LIDAR sensor and onboard LIDAR in the data capture system were validated against drone tracking data representing the pseudo-ground truth. This study may have been affected by some known errors in drone tracking, as drone tracking results are considered as pseudo ground truth and and error calculation as shown in Table 1. Some known errors are due to non-planar ground, which is assumed to be planar in drone tracking.

In future work, one thing the infrastructure based system can take advantage of is the fusion of multiple LIDAR point cloud from different LIDAR positioned at different view point. This will not only make the point cloud dense but also it will have points of same object from different angles, that will definitely improve the detection and tracking, and eventually safety metrics calculation. In addition, GPS data could be a more reliable source for ground truth rather than drone camera (as later is subjected to spatial error due to non-planar ground), which can be done by installing a GPS sensor in the follower vehicle as well. Otherwise, inertial measurement units(IMUs) can be installed in both the vehicles instead of GPS sensors.

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Acknowledgments

Authors are grateful to the Institute of Automated Mobility (IAM) for generously funding this work and Maricopa County Department of Transportation (MCDOT) for enabling infrastructural resources required for this work. HY, and SD acknowledge the support from National Science Foundation (grant nos. 1624842, 1839485, 1933742, 2137295) Industry University Cooperative Research Center (IUCRC) for Efficient Vehicles and Sustainable Transportation Systems (EVSTS) at Arizona State University.

ISSN 0148-7191